Enhancing Small Language Models for Co-Creative Mathematical Storytelling using RAG

Background

Recent work such as *Mathemyths (CHI 2024)* demonstrated that large language models (LLMs) can support children's learning of mathematical language through interactive co-creative storytelling. However, such systems rely on large GPU-based models, limiting practical deployment in low-resource educational environments.

This project aims to replicate and extend this idea using **small, CPU-runnable language models (e.g., Gemma-class ~2B models)** and investigate how retrieval augmentation can enhance their ability to generate **coherent, mathematically accurate long stories**.

Project Objectives

Students will:

1. Implement a **co-creative storytelling system** where an AI and user collaboratively build mathematical stories.
2. Use a **small open-source language model** that runs efficiently on CPU.
3. Investigate **coherence degradation** in long multi-turn narratives (10–20+ turns).
4. Design and implement a **Retrieval-Augmented Generation (RAG)** pipeline using curated Grade 1–6 mathematics content.
5. Evaluate improvements in narrative coherence and mathematical correctness with and without RAG.

Technical Scope

Model:

- Lightweight LLM (quantized if needed) runnable on CPU.

Knowledge Base:

- Structured Grade 1–6 mathematics content:
  - Arithmetic, fractions, geometry basics, ratios, word problems
  - Mathematical vocabulary and example explanations

RAG Pipeline:

- Vector database (e.g., FAISS)
- Lightweight embedding model
- Context-aware retrieval integrated into prompts

Research Questions

- How does long-form narrative coherence degrade in small language models?
- Can RAG significantly improve:
    - Mathematical accuracy?
    - Concept integration?
    - Logical and narrative consistency?
- What trade-offs exist between model size, retrieval quality, and fluency?

Evaluation Plan

1. Narrative Coherence

- Topic drift analysis
- Entity consistency tracking
- Human scoring (flow, consistency, engagement)

2. Mathematical Accuracy

- Correctness of embedded math concepts
- Error rate in generated word problems
- Concept usage coverage

3. Comparative Study

- Base small model vs. RAG-enhanced model

Expected Outcomes

- A CPU-deployable AI storytelling prototype
- Quantitative analysis of long-story coherence in small models
- Demonstration of RAG as a capability amplifier for small LLMs
- Final technical report and live system demo